

Causal Inference Crash Course:

Part 10 — Conformal Inference

Julian Hsu

Causal Inference Series

1. Foundations
2. Causal Models
3. Inference
4. Best Practices
5. Heterogeneous Treatment Effects
6. Panel Models
7. Regression Discontinuity
8. Surrogate Models
9. Arguable Validation
10. **Conformal Inference** ← **this deck**

Overview

- **Conformal inference** turns *any* model's point prediction into a **set** with a **finite-sample, distribution-free coverage guarantee** — it needs only **exchangeability**, not a distribution and not a correctly specified model.
- Why we care: honest uncertainty for the black-box predictors we actually use — and, the payoff, **valid causal inference with a single treated unit** (synthetic control), where classical standard errors don't apply.
- We cover:
 - the idea: **split conformal** and its coverage guarantee
 - the one assumption — **exchangeability** — and what breaks it
 - two fixes when it fails: **moving blocks** and **weighting**
 - the causal payoff: conformal inference for **synthetic controls**
- Assumes the earlier **Inference** and **Panel Models** decks.

The Idea

What we want

- Classical prediction intervals lean on a **model being right** — Wald needs a correct parametric form and large samples; Bayes needs a correct likelihood.
- **Conformal** asks for far less: a set $\mathcal{C}(x)$ with

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha$$

that holds in **finite samples**, for **any** data distribution, wrapped around **any** predictor.

- Running example: predict a new customer's next-quarter purchases and give an interval that really covers 90% of the time — whatever model produced the point prediction.

The one assumption: exchangeability

- The observations $(X_1, Y_1), \dots, (X_{n+1}, Y_{n+1})$ are **exchangeable**: their joint distribution is unchanged by reordering them. (i.i.d. is the familiar special case.)
- That is the *whole* assumption — no distributional form, no correctly specified model. Everything downstream is a consequence of it.
- It is also exactly what **fails** for time series and panels — we will come back to that.

Conformity scores

- Pick a **nonconformity score** $s(x, y)$ — "how strange is this pair, to the model?" Larger means a worse fit.
- Regression default: the absolute residual of any fitted $\hat{f}$,

$$s(x, y) = |y - \hat{f}(x)|.$$

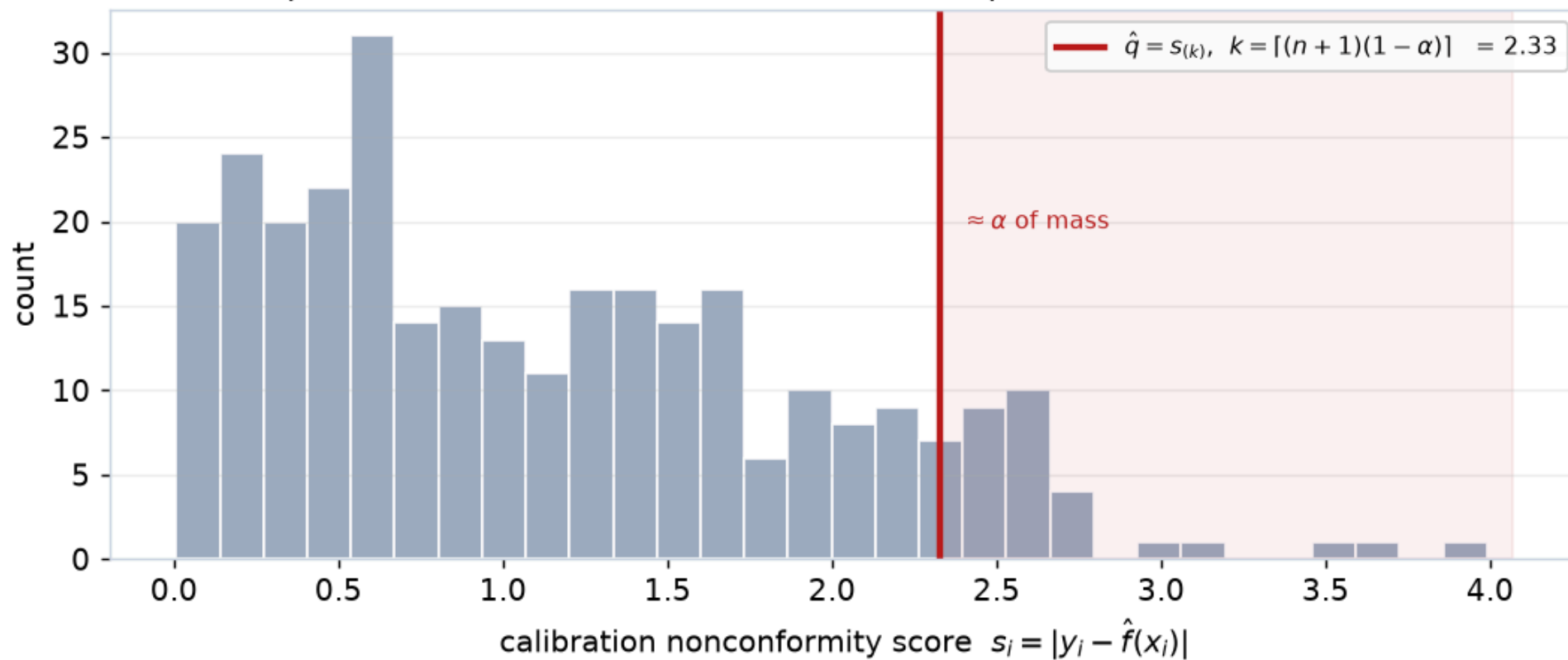
- The guarantee holds for **any** score and **any** model; a *good* score only makes the resulting set **tighter**, never invalid.

Split conformal — the recipe

For miscoverage level α , split the data into a **training** fold and a **calibration** fold of size n :

1. **Fit** $\hat{f}$ on the training fold only.
2. **Score** the calibration set: $s_i = s(X_i, Y_i)$, $i = 1, \dots, n$.
3. **Quantile**: $\hat{q} = s_{(k)}$, the k -th smallest score, with $k = \lceil (n + 1)(1 - \alpha) \rceil$.
4. **Predict**: $\mathcal{C}(x) = \{ y : s(x, y) \leq \hat{q} \}$.

Split conformal: threshold at the $(1-\alpha)$ quantile ($n=300$, $\alpha=0.1$)



The coverage guarantee

- **Theorem** (Vovk et al. 2005; Lei et al. 2018). If the data are **exchangeable**, split conformal satisfies

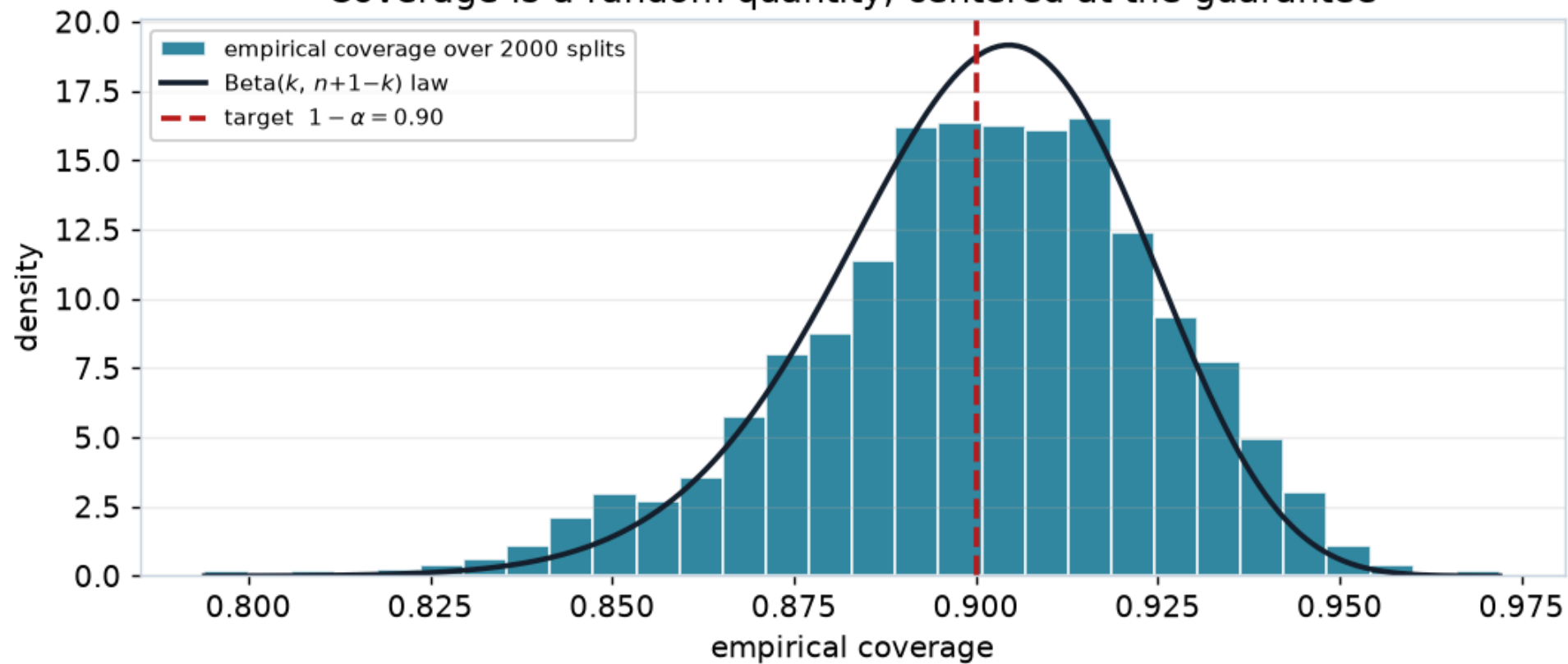
$$1 - \alpha \leq \mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \leq 1 - \alpha + \frac{1}{n+1}.$$

- **Finite-sample, distribution-free, model-free.** The upper bound needs only that scores have no ties.
- The model can be **arbitrarily wrong** — coverage still holds. A bad model simply yields an honestly **wide** set.

Why it works

- Insert the unseen test score as an $(n+1)$ -th point. By exchangeability its **rank is uniform** on $\{1, \dots, n+1\}$, so it exceeds the k -th calibration value only about α of the time — which is exactly the miscoverage event.
- So coverage is a **property of the procedure**, not of any one interval: over repeated draws it follows a $\text{Beta}(k, n+1-k)$ law centered at $1 - \alpha$.

Coverage is a random quantity, centered at the guarantee



The p-value view

- Equivalent framing (Shafer & Vovk 2008): conformal is **hypothesis testing turned into prediction**. For each candidate label y ,

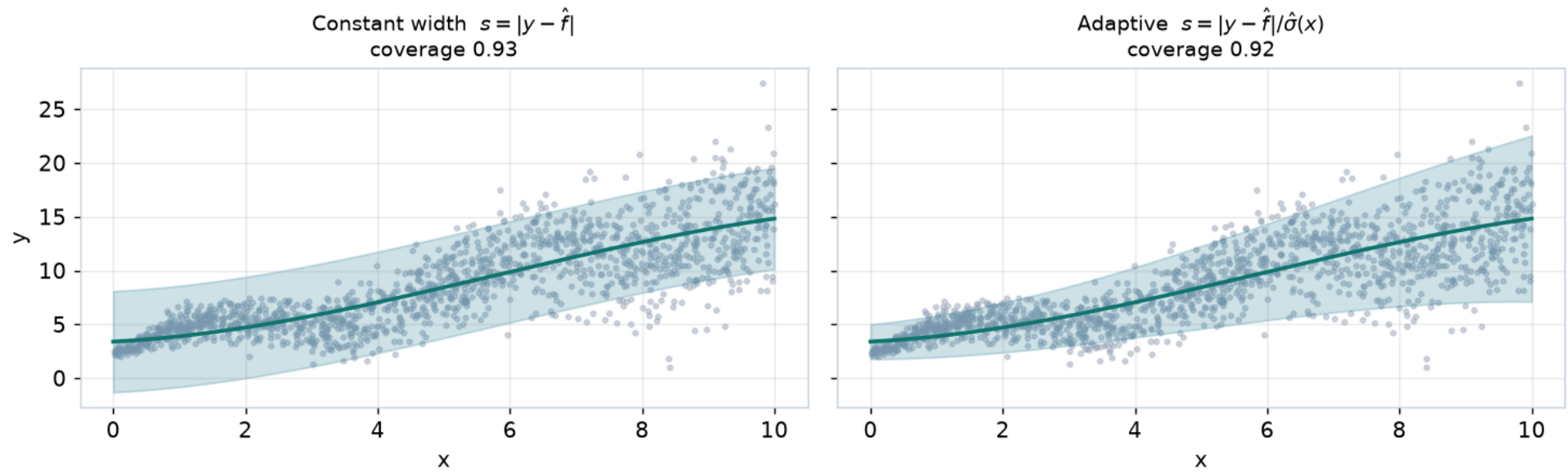
$$p(y) = \frac{1 + \#\{i : s_i \geq s(x, y)\}}{n + 1}, \quad \mathcal{C}(x) = \{y : p(y) > \alpha\}.$$

- Keep the labels you **cannot reject** at level α .
- This is the view we will use for synthetic control — where the "label" becomes a hypothesized treatment effect.

Adaptivity — and its limit

- The guarantee is **marginal**: it averages over draws. It is **not conditional** — some regions can be under-covered and others over-covered, and exact conditional coverage is impossible distribution-free.
- Fix with a smarter score, not a new guarantee: normalize by a local scale $s = |y - \hat{f}(x)| / \hat{\sigma}(x)$, or use **conformalized quantile regression** (Romano, Patterson & Candès 2019).

Same 90% marginal coverage, very different local behavior

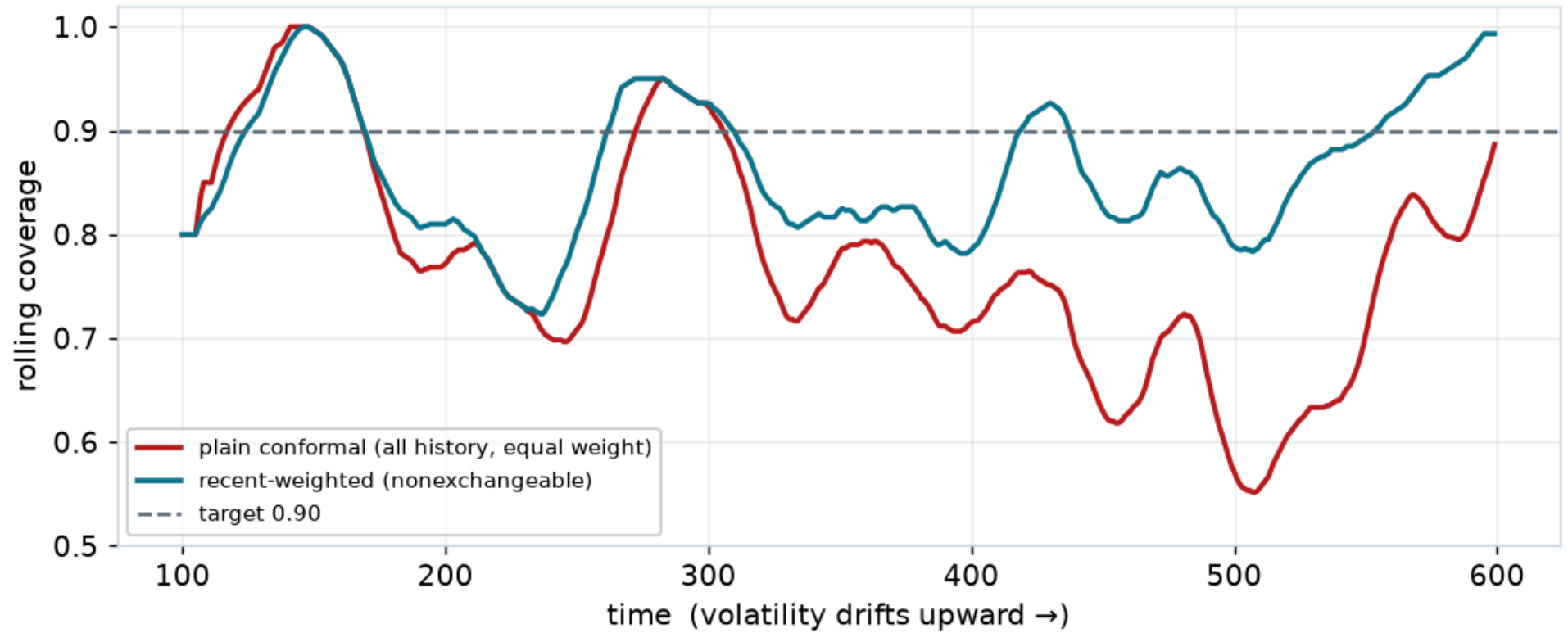


When Exchangeability Fails

Time and treatment break exchangeability

- Panels and time series are **serially dependent** and often **drift** — reorder them and the joint law changes, so exchangeability fails.
- Worse for causal work: the null holds only **before** treatment, and the post-period is exactly what we suspect differs.
- Plain conformal then **under-covers** — its interval is calibrated on a stale past that no longer resembles the present.

Under drift, exchangeability breaks and plain conformal under-covers



Fix A — permute blocks, not points

- Keep **contiguous blocks** of residuals intact and permute the block **positions** — the moving-block idea borrowed from the block bootstrap.
- Local dependence rides along *inside* each block; only its placement is randomized, so the null datasets keep the real time structure.
- **Exact** under exchangeability, **approximately valid** under stationarity and weak dependence — the block length must exceed the dependence range (Chernozhukov, Wüthrich & Zhu 2018, 2021).

Fix B — weight what belongs

- **Nonexchangeable conformal** (Barber, Candès, Ramdas & Tibshirani 2023): attach fixed weights w_i and take **weighted quantiles** of the scores — for time series, weight **recent** points more.
- The guarantee degrades gracefully:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C}(X_{n+1})) \geq 1 - \alpha - \Delta,$$

where Δ is a weighted **distance from exchangeability**; it vanishes when the up-weighted points really do resemble the target.

- The weights must be **fixed a priori** — not learned from the scores.

Two relaxations of one idea

- **Blocks** — the *discrete* relaxation: glue dependent points together so a permutation can't shatter the local structure.
- **Weights** — the *continuous* relaxation: down-weight the points that don't belong with the target.
- Both **restore an approximate symmetry** the raw data lacks — just enough for the uniform-rank argument to (nearly) go through.

Conformal Inference for Synthetic Controls

The CWZ recast

- Chernozhukov, Wüthrich & Zhu (2021) glue two familiar problems together: causal inference = **counterfactual prediction** + **a test for a structural break**.
- Predict the treated unit's *untreated* path (synthetic control, DiD, factor models...), then ask whether the **post-treatment residuals break** from the pre-treatment ones.
- The pitch: **inference that doesn't care how good your synthetic control is** — validity comes from the permutation, not from a correct model.

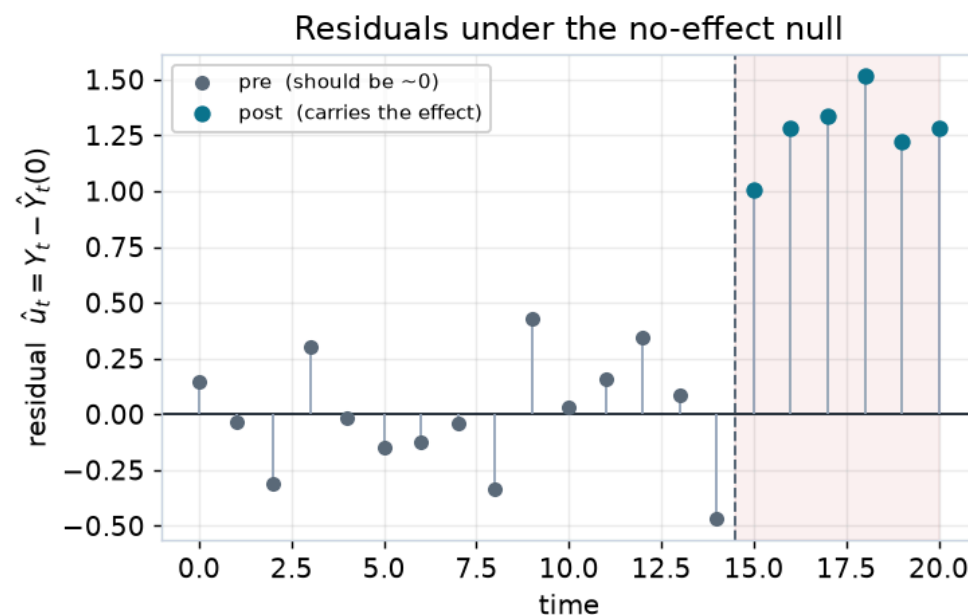
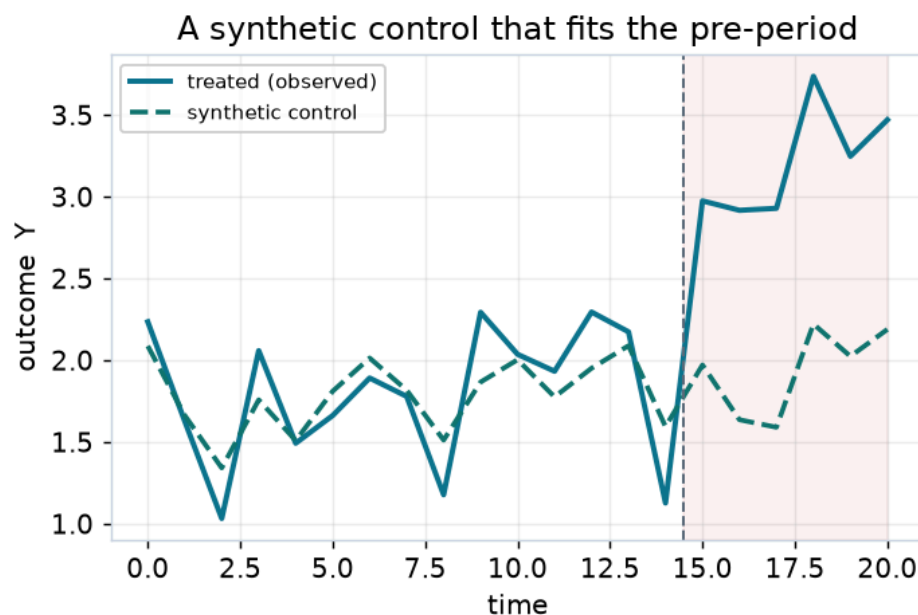
Setup — we predict the counterfactual

- One treated unit Y_{1t} , a donor pool Y_{0t} , pre-periods $t \leq T_0$ and post-periods $t > T_0$. Fit a **synthetic control** $\hat{Y}_{1t}(0)$ (from the Panel Models deck).
- The **gap** $\hat{u}_t = Y_{1t} - \hat{Y}_{1t}(0)$ is the residual. Under *no effect*, that residual process has **no break** at T_0 .
- The residuals over time are serially dependent — so this is exactly the non-exchangeable case, handled by **moving-block** permutations.

Step 1 — impose the null, read the residuals

- Test a sharp null on the post-period effect — here a constant effect
 $H_0 : \tau_t = \theta$. Subtract it from the **post outcomes only**, refit over the **whole timeline**, and take residuals for every t :

$$\tilde{Y}_{1t} = Y_{1t} - \theta \cdot \mathbf{1}\{t > T_0\}, \quad \hat{u}_t = \tilde{Y}_{1t} - \hat{Y}_{1t}(0)$$



Step 2 — a statistic and a moving-block permutation

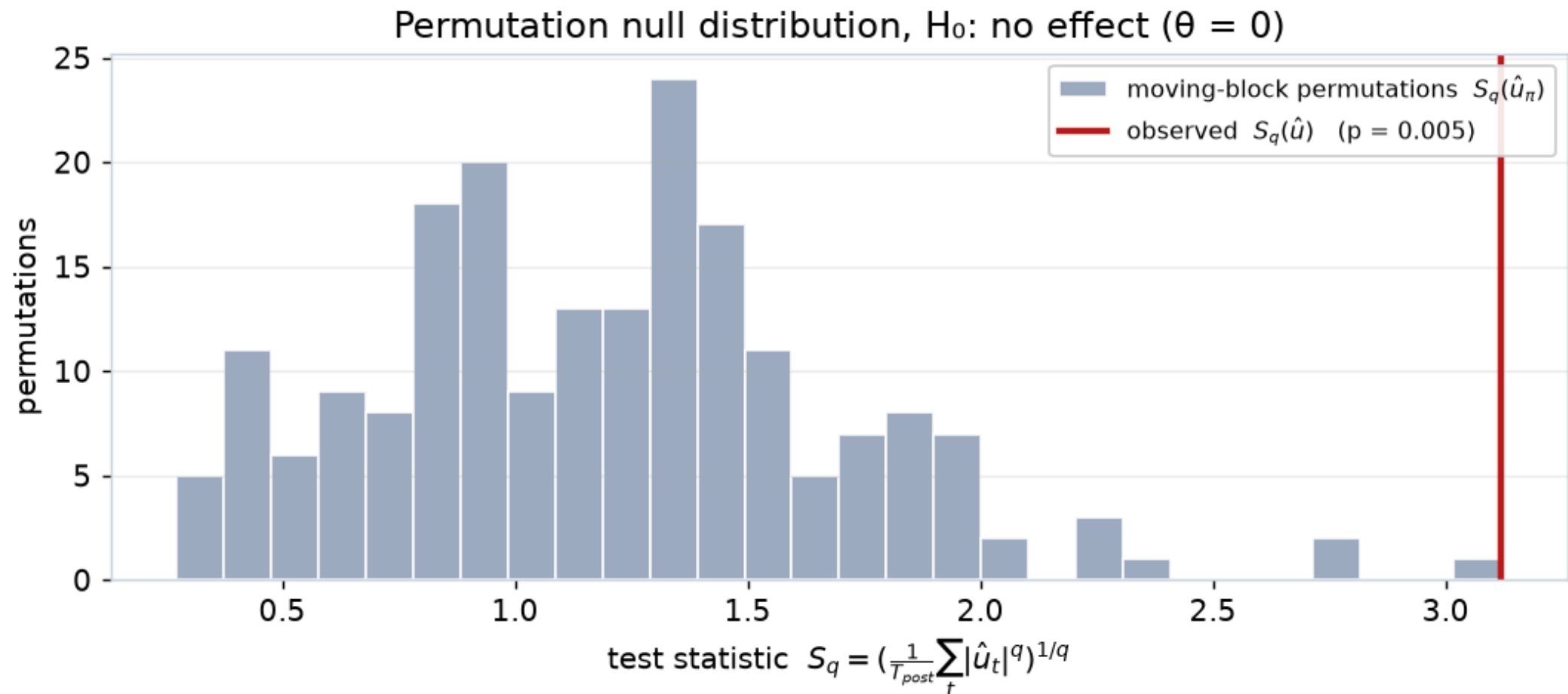
- Summarize the post-period residuals with a single number ($q = 1$ = mean absolute residual):

$$S_q(\hat{u}) = \left(\frac{1}{T_{post}} \sum_{t > T_0} |\hat{u}_t|^q \right)^{1/q}$$

- Compare S_q to its distribution under **moving-block permutations** of the residual sequence — the serial-dependence-robust scheme from the last section.
- `panelib`'s `conformal_inf` implements exactly this.

Step 3 — the permutation p-value

$$p = \frac{1}{|\Pi|} \sum_{\pi \in \Pi} \mathbf{1}\{S_q(\hat{u}_\pi) \geq S_q(\hat{u})\}, \quad \text{reject } H_0 \text{ if } p \leq \alpha.$$

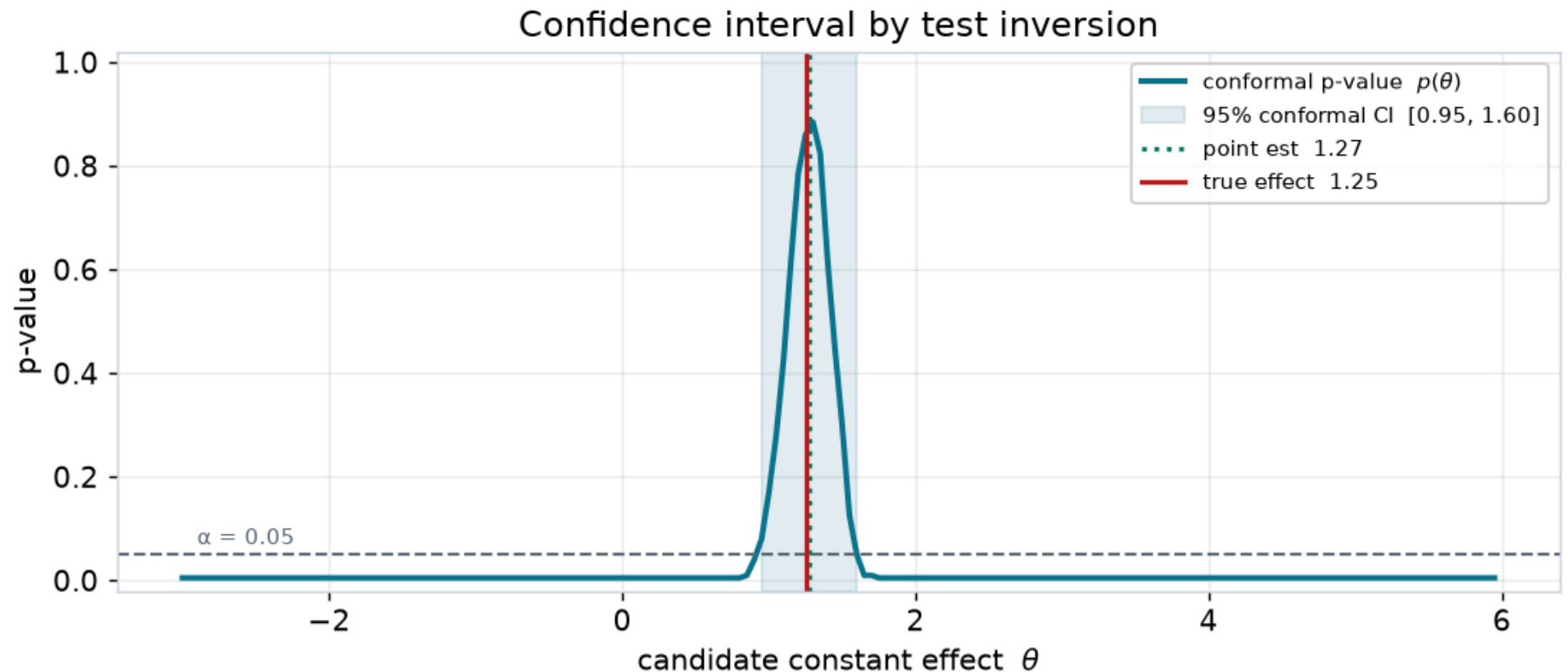


- **Careful:** this permutes residuals **across time** for one treated unit — not the Abadie placebo test, which permutes treatment **across units**.

Step 4 — confidence interval by test inversion

- Repeat the test over a grid of candidate effects θ and keep every value you **cannot reject**:

$$CI_{1-\alpha} = \{ \theta : p(\theta) > \alpha \}$$



- The point estimate sits where $p(\theta)$ peaks; the CI is the set of effects consistent with "no break."

Why this matters

- Works with **one treated unit and few pre-periods** — and is **exact** under exchangeability, **robust** to serial dependence and to a **misspecified** counterfactual (validity comes from residual symmetry, not a correct model).
- The fine print: under dependence the guarantee is approximate, and it needs the counterfactual's **error to be stable over time** — a model whose bias *trends* into the post-period breaks a symmetry no permutation can restore.
- Resolution matters too: with very few post-periods the permutation p-value is coarse — in the Panel deck's simulations (2 post-periods) coverage ran **80–85%**, not 95%.
- Contrast Abadie placebo inference: its p-value resolution is only $1/(J + 1)$ and it needs a large donor pool.
- This is the inference the **Panel Models** deck used to put DiD, SC, and SDID on the same footing.

Conclusion

- Conformal inference buys an **honest, finite-sample guarantee** from **one assumption you can actually reason about** — exchangeability — and degrades gracefully when it fails.
- Reach for it to **wrap black-box predictors**, and — the causal payoff — to do inference where classical SEs can't: **a single treated unit over time**.
- The price: coverage is **marginal**, not conditional; and under dependence you must **block** or **weight** to keep the guarantee.
- Start from the guarantee you can defend, then pick the score and the permutation that fit your data.

Appendix

Appendix: split vs. full conformal

- **Split (inductive)** conformal — the recipe here — spends part of the data on a calibration fold. Cheap (one fit), but the interval is random in that split and a bit wider.
- **Full** conformal refits for every candidate label — exact but expensive; **CV+ / jackknife+** (Barber et al. 2021) recover most of the efficiency without a held-out split.
- Same guarantee; the choice is **data efficiency vs. compute**.

Appendix: the coverage law

- Conditional on the calibration draw, the coverage of split conformal is a **random variable**:

$$\mathbb{P}(Y_{n+1} \in \mathcal{C} \mid \text{calibration}) \sim \text{Beta}(k, n + 1 - k), \quad k = \lceil (n + 1)(1 - \alpha) \rceil.$$

- Its mean is $\geq 1 - \alpha$; the spread shrinks like $1/\sqrt{n}$ — the middle histogram made this concrete.
- The upper bound $1 - \alpha + \frac{1}{n+1}$ needs the scores to have **no ties** (continuous scores, or a random tie-break).

Appendix: another route — prediction intervals

- Conformal is not the only distribution-aware option for synthetic control.
- **Cattaneo, Feng & Titiunik (2021)** build **prediction intervals** that separate two error sources: the in-sample **weight-construction** error and the out-of-sample **stochastic** error, with finite-sample guarantees.
- Rule of thumb: **conformal** tests a sharp null by permutation and inverts to a CI; **prediction intervals** model the two error sources directly. Different assumptions, complementary diagnostics.

Appendix: conformal for treatment effects, beyond panels

- **Individual treatment effects** — Lei & Candès (2021): conformal intervals for $Y(1) - Y(0)$ under unconfoundedness, using *weighted* conformal to bridge the treated/control covariate shift. The natural companion to the HTE deck.
- **Covariate shift** — Tibshirani, Barber, Candès, Ramdas (2019): weighted conformal stays valid when the test point comes from a different covariate distribution with a known (or estimated) likelihood ratio.
- **Drift in production** — Gibbs & Candès (2021): *adaptive* conformal updates the level α_t online to keep realized coverage on target under distribution shift — the monitoring use case for deployed models.

Literature Review of Related Papers

- **Conformal prediction — foundations**

- Vovk, Gammerman, Shafer (2005). *Algorithmic Learning in a Random World*. Springer — the founding monograph. [link](#)
- Shafer & Vovk (2008). "A Tutorial on Conformal Prediction." *JMLR* 9, 371–421. [link](#)
- Lei, G'Sell, Rinaldo, Tibshirani, Wasserman (2018). "Distribution-Free Predictive Inference for Regression." *JASA* 113(523). [link](#)
- Angelopoulos & Bates (2023). "A Gentle Introduction to Conformal Prediction and Distribution-Free Uncertainty Quantification." *Found. & Trends ML*. [link](#)

Literature Review (continued)

- **Adaptivity & beyond exchangeability**

- Romano, Patterson, Candès (2019). "Conformalized Quantile Regression." *NeurIPS*. [link](#)
- Barber, Candès, Ramdas, Tibshirani (2021). "Predictive Inference with the Jackknife+." *Ann. Statist.* 49(1). [link](#)
- Barber, Candès, Ramdas, Tibshirani (2023). "Conformal Prediction Beyond Exchangeability." *Ann. Statist.* 51(2), 816–845. [link](#)
- Tibshirani, Barber, Candès, Ramdas (2019). "Conformal Prediction Under Covariate Shift." *NeurIPS*. [link](#)
- Gibbs & Candès (2021). "Adaptive Conformal Inference Under Distribution Shift." *NeurIPS*. [link](#)

Literature Review (continued)

- **Conformal inference for causal panels / synthetic control**

- Chernozhukov, Wüthrich, Zhu (2018). "Conformal Inference Methods for Predictive ML with Dependent Data." *COLT (PMLR 75)*. [link](#)
- Chernozhukov, Wüthrich, Zhu (2021). "Conformal Inference for Counterfactual and Synthetic Controls." *JASA* 116(536). [link](#)
- Cattaneo, Feng, Titiunik (2021). "Prediction Intervals for Synthetic Control Methods." *JASA* 116(536). [link](#)
- Firpo & Possebom (2018). "Synthetic Control Method: Inference, Sensitivity Analysis and Confidence Sets." *J. Causal Inference* 6(2). [link](#)
- Lei & Candès (2021). "Conformal Inference of Counterfactuals and Individual Treatment Effects." *JRSS-B* 83(5). [link](#)

- **Companion code:** `panelib.py` (`conformal_inf`) in [statanomics](#).

